

Isabel Rosa Marie Medlock
PhD Candidate at Yale; Visitor at CfA

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EDUCATION

Yale University	New Haven, CT
<i>Astronomy PhD Candidate, Thesis Advisor: Prof. Daisuke Nagai</i>	<i>Aug 2021 – May 2027</i>
Princeton University	Princeton, NJ
<i>A.B. in Astrophysical Sciences; Certificates in Applications of Computing and Russian Language</i>	<i>Sep 2017 – Jun 2021</i>

RESEARCH INTERESTS

I study how baryons shape galaxy evolution and bias cosmological measurements through a computational and modeling lens. I am specifically interested in cold stream gas accretion, the baryon cycle in the circumgalactic medium, and fast radio bursts as novel cosmological probes.

PUBLICATIONS

14 publications: 6 first-author (5 refereed, 1 submitted), 6 refereed co-author, 2 non-refereed.

AWARDS & GRANTS

NSF ACCESS Computing Grant (Science-PI, Lead Author):	<i>2025</i>
312.5k Stampede3 node hours (\$68.5k)	
NSF ACCESS Computing Grant (Science-PI, Lead Author):	<i>2024</i>
300k Stampede3 node hours (\$62k)	
Yale Dean's Fund for Colloquia and Symposia, for Yale Cosmology Seminar (\$2,700)	<i>2024 – 2026</i>
Travel Grants: Yale Graduate Student Assembly Conference Travel Fellowship (\$2,400; 2023, 2024, 2025), AAS (\$2,000; 2023, 2024), APS Division of Astrophysics (\$300; 2023), SACNAS NDiSTEM (\$1,000; 2022)	<i>2022 – 2025</i>
Dean's Emerging Scholars Research Award, Yale (\$2,000)	<i>2022</i>

TALKS

Invited:

Galaxies Across Cosmic Time in Paso Robles, CA	<i>Oct 2026</i>
Baryons @ Princeton at Princeton University	<i>May 2026</i>
Astronomy Seminar at University of Connecticut	<i>Mar 2026</i>
33rd Texas Symposium: FRB Parallel Session in Tempe, AZ	<i>Dec 2025</i>
ITC Luncheon at Harvard-Smithsonian Center for Astrophysics	<i>Oct 2025</i>
Borthakur Group Meeting at Arizona State University (virtual)	<i>Oct 2025</i>
FRB Journal Club at Max Planck Institute for Radio Astronomy (virtual)	<i>Apr 2025</i>
Astro Lunch at University of Washington	<i>Feb 2025</i>

Selected Contributed:

Summary (2020–present): 26 talks, 4 posters	
Filaments in the Multi-Scale Universe at IPMU, Tokyo, Japan	<i>Oct 2026</i>

Dynamic Cosmic Structures in Hangzhou, China	<i>Oct 2026</i>
Cosmic Ecosystems at Perimeter Institute, Waterloo, Canada	<i>Aug 2025</i>
Fast Radio Burst 2025 in Montreal, Canada	<i>Jul 2025</i>
Cosmology and Galaxy Astrophysics with Simulations and ML at CCA	<i>Dec 2024</i>
Fast Radio Burst 2024 in Khao Lak, Phang Nga, Thailand	<i>Nov 2024</i>
Baryons in the Universe 2024 at Kavli IPMU, Kashiwa, Japan	<i>Apr 2024</i>

STUDENT MENTORING

Summary: 1 graduate and 1 undergraduate mentee; 1 co-authored publication

Donglin Wu, Yale Undergraduate Student *Aug 2025 – present*

- FRB Host Halo Dispersion Measures in CAMELS-TNG50, paper in preparation.

Chloe Neufeld, Yale Graduate Student *Sep 2023 – Dec 2024*

- Feedback energetics in CAMELS, second author on Medlock et al. 2025, ApJ 980, 61.

PROFESSIONAL SERVICE

Journal Referee (ApJ, OJA, A&A; 3 Manuscripts)	<i>2026 – present</i>
First-Year Astronomy Buddy (FAB) Program Mentor	<i>Aug 2024 – May 2026</i>
SACNAS Yale Chapter Co-President, Secretary, and Treasurer	<i>Jan 2023 – present</i>
Yale Cosmology Seminar Co-Organizer	<i>Sep 2023 – present</i>
Yale Astronomy Siblings Mentor	<i>Sep 2022 – present</i>
Astronomy Climate and Diversity Committee Member	<i>Sep 2022 – present</i>

OUTREACH

LFOP Spanish Language Night Creator and Organizer	<i>Feb 2023 – present</i>
Leitner Family Observatory and Planetarium (LFOP) Presenter for School Visits	<i>Sep 2022 – present</i>
Science in the News Presenter	<i>Oct 2022 – May 2023</i>

TEACHING EXPERIENCE

Yale Pathways Instructor for Local High School Students	<i>Jul 2025, Jul 2026</i>
Eduexplora Yale Summer Session Instructor	<i>Jul 2023</i>
Teaching Fellow, Yale: Galaxies and the Universe; Frontiers and Controversies in Astrophysics; Intro to Relativity and Black Holes; Planets and Stars	<i>Sep 2021 – May 2023</i>

REFERENCES

Daisuke Nagai (PhD Advisor): Professor of Physics and of Astronomy, Yale University

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Frank van den Bosch (Collaborator): Professor of Astronomy and Physics, Yale University

Email: frank.vandenbosch@yale.edu, Phone: +1 203-432-0196

Volker Springel (Collaborator): Director, Max Planck Institute for Astrophysics

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Isabel Medlock: Publications

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FIRST AUTHOR

Published and Submitted:

- [6] **Medlock, I.**, and Nagai, D., “Beyond Feedback: Disentangling Baryonic Effects with tSZ x FRB Cross-Correlations”, submitted to *ApJ*
- [5] **Medlock, I.**, Nagai, D., Mandelker, N., Springel, V., van den Bosch, F., Zinger, E., and Chiang, B., “Statistical Properties of Cold Streams in Massive Star-Forming Halos in IllustrisTNG-50”, 2025, *ApJ*, 1000(2), 222
- [4] **Medlock, I.**, Nagai, D., Anglés-Alcázar, D., and Gebhardt, M., “Constraining the Effect of Baryonic Feedback on the Matter Power Spectrum with Fast Radio Bursts”, 2025, *ApJ*, 983(1), 46
- [3] **Medlock, I.**, Neufeld, C., Nagai, D., Anglés-Alcázar, D., Genel, S., Oppenheimer, B., Singh, P., and Villaescusa-Navarro, F., “Quantifying Baryonic Feedback on Warm-Hot Circumgalactic Medium in CAMELS Simulations”, 2025, *ApJ*, 980(1), 61
- [2] **Medlock, I.**, Nagai, D., Singh, P., Oppenheimer, B., Anglés-Alcázar, D., and Villaescusa-Navarro, F., “Probing the Physics of the Circum-galactic Medium using Fast Radio Bursts: Insights from CAMELS”, 2024, *ApJ*, 967(1), 32
- [1] **Medlock, I.**, and Cen, R., “Dispersion Measure Distributions of Fast Radio Bursts Due to the Intergalactic Medium”, 2021, *MNRAS*, 502(3), 3664–3669

In Preparation:

- [9] **Medlock, I.**, Gallaraga-Espinosa, D., Nagai, D., et al, “Gaseous Line-Mass II: Observing Cosmic Filaments with the tSZ and kSZ Effects”, in preparation
- [8] **Medlock, I.**, and Nagai, D., “FRB Dispersion Measure Statistics from the Baryon Pasting Halo Model: Feedback Sensitivity and Validity Limits”, in preparation
- [7] **Medlock, I.**, Nagai, D., et al, “Cold Accretion Streams Penetrate Deeper with Increased Resolution: First Results from Zoom-In Simulations”, in preparation

REFEREED CO-AUTHOR

Published and Submitted:

- [6] Sims, X., Anglés-Alcázar, D., Oh, B. K., Nagai, D., Mercedes-Feliz, J., **Medlock, I.**, Ni, Y., Lovell, C., and Villaescusa-Navarro, F., “CAMELS Environments: The Impact of Local Neighbours on Galaxy Evolution across the SIMBA, IllustrisTNG, ASTRID, and Swift-EAGLE Simulations”, 2026, *MNRAS*, 551(4), 1529
- [5] Gebhardt, M., Anglés-Alcázar, D., Genel, S., Nagai, D., Oh, B. K., **Medlock, I.**, Mercedes-Feliz, J., Sutherland, S., Lee, M., Sims, X., Lovell, C., Spergel, D., Davé, R., Schaller,

M., Schaye, J., and Villaescusa-Navarro, F., “Cosmological Back-Reaction of Baryons on Dark Matter in the CAMELS Simulations”, 2026, *MNRAS*, 547(4), 525

[4] Oppenheimer, B., Voit, M., Bahé, Y., Battaglia, N., Bregman, J., Burchett, J., Eckert, D., Faerman, Y., Gibson, J., Hummels, C., **Medlock, I.**, Nagai, D., Putman, M., Qu, Z., Sun, M., Werk, J., and Zhang, Y., “The Descriptive Parametric Model I: Gaseous Profiles for Galaxies, Groups, and Clusters”, 2025, *MNRAS*, 543(3), 2649–2669 (*FRB Model lead*)

[3] Zhang, Z., Nagamine, K., Oku, Y., Lee, K.-G., Fukushima, K., Tomaru, K., Zhang, B., **Medlock, I.**, and Nagai, D., “Probing the Cosmic Baryon Distribution and the Impact of AGN Feedback with FRBs in CROCODILE Simulation”, 2025, *ApJ*, 993(2), 162

[2] Leung, C., Simha, S., **Medlock, I.**, Nagai, D., Masui, K., Kahinga, L., Lanman, A., Andrew, S., Bandura, K., Curtin, A., Gaensler, B., Gusinskaia, N., Joseph, R., Lazda, M., Mas-Ribas, L., Meyers, B., Nimmo, K., Pearlman, A., Prochaska, J. X., Sammons, M., Shin, K., Smith, K., and Wang, H., “Stellar Mass-Dispersion Measure Correlations Constrain Baryonic Feedback in Fast Radio Burst Host Galaxies”, 2025, *ApJL*, 991(1), L25 (*Simulation analysis lead*)

[1] Lau, E., Nagai, D., Bogdán, Á., **Medlock, I.**, Oppenheimer, B., Battaglia, N., Anglés-Alcázar, D., Genel, S., Ni, Y., and Villaescusa-Navarro, F., “X-raying CAMELS: Constraining Baryonic Feedback in the Circum-Galactic Medium with the CAMELS Simulations and eRASS X-ray Observations”, 2025, *ApJ*, 984(2), 190

In Preparation:

[8] Wu, D., **Medlock, I.**, Nagai, D., et al. “FRB Host DM as a Constraint of Baryonic Feedback: Insights from CAMELS”, in preparation (*Primary mentor*)

[7] Gallaraga-Espinosa, D., Nagai, D., **Medlock, I.**, et al., Gaseous Line-Mass I: A Unifying Descriptor of Cosmic Filaments, in preparation (*Analytic filament model lead*)

NON-REFEREED CO-AUTHOR

[2] Oppenheimer, B., Nagai, D., Lau, E., Singh, P., Butler Contreras, A., Gluck, N., Dorigo Jones, J., **Medlock, I.**, and Villaescusa-Navarro, F., “A Multi-Wavelength, Multi-Model Exploration of How Feedback Disrupts Gaseous Atmospheres”, 2022, *Bulletin of the AAS*, 54(1)

[1] Singh, P., Nagai, D., Oppenheimer, B., Lau, E., Gluck, N., and **Medlock, I.**, “Galactic Gaseous Halos: Mini-Clusters Disrupted by Feedback”, 2022, *Galactic Atmospheres*